Water Leak Detection: Scenario Catalogue

EPANET Net3 benchmark (92 junctions, 2 reservoirs, 3 tanks, 117 pipes, 2 pumps) | 24 h extended-period simulation | WNTR pressure-dependent demand

1. Purpose

Semester 1 tested only two leak cases (node 15 and node 115). This catalogue defines **41 distinct scenarios** (S00-S40) so the detector and localizer are tested on many different situations. Every scenario has the same end goal: **find the node where the leak occurred**. Scenarios differ in where the leak is, how big it is, when it starts, how many leaks exist, and what other disturbances (heavy flow, blocked pipes, pump trips, sensor faults) are present at the same time.

2. How each scenario is built

Each scenario is one entry in a scenario list that `simulation/generate_dataset.py` loops over. It writes `<name>_pressures.csv` and `<name>_flows.csv` plus a ground-truth label (the true leak node list). Leaks use `node.add_leak(wn, area, start_time, end_time)`. Blockages use a closed pipe or a large minor-loss factor. Pump trips use a pump status change. Heavy flow uses an extra demand at a junction. Demand scaling uses the global demand multiplier. Sensor noise and sparse sensors are applied to the CSVs after simulation, so no extra simulation is needed.

3. What counts as success

Top-1: the highest-ranked node is the true leak node. **Top-5**: the true node is among the five highest-ranked. For multi-leak scenarios every true node must appear in the Top-K. For no-leak scenarios (S00, S28, S30, S32, S33) the system must raise **no leak alarm**; these measure the false-alarm rate. Levels (Easy to Very hard) are my estimates from the physics, not measured results; they will need to be checked once the scenarios are simulated.

4. Coverage summary

Group	Scenarios	What it varies
A. Control cases	S00 - S00 (1)	No-leak reference
B. Single leak - different places in the network	S01 - S14 (14)	Leak location and node type (dead-end, hub, pump, tank, elevation, demand)
C. Same location, different leak size	S15 - S18 (4)	Leak size at a fixed node
D. Different timing of the leak	S19 - S23 (5)	Start time, duration and growth
E. Multiple leaks at the same time	S24 - S27 (4)	Number and arrangement of leaks
F. Blocked / heavy flow	S28 - S34 (7)	Blockages, heavy flow and pump faults (with and without a leak)
G. Operating-condition and data-quality variations	S35 - S40 (6)	Tank level, seasonal demand, sensor noise, sparse and faulty sensors

Scenario Catalogue

Leak area is the orifice area in square metres. 'Timing: 12:00 on' means the leak starts at hour 12 and stays until the end of the 24 h run. Node numbers are Net3 junction IDs.

A. Control cases (no leak)

ID	Scenario	Leak node(s)	Area (m2)	Timing	Setup (how to simulate)	Why it is unique / challenge	Level
S00	Normal operation	None	-	24 h	Baseline run, no events.	False-alarm check. System must report NO leak and no location.	Easy

B. Single leak - different places in the network (ground truth = the leak node)

ID	Scenario	Leak node(s)	Area (m2)	Timing	Setup (how to simulate)	Why it is unique / challenge	Level
S01	Dead-end, low demand	15	0.05	12:00 on	Existing scenario. Outer branch, demand ~1 gpm.	Leak is the only thing moving flow at this node; clear signal.	Easy
S02	Central 4-way hub	115	0.03	12:00 on	Existing scenario. Degree-4 junction, demand ~52 gpm.	Pressure drop spreads to 4 neighbours, so localisation is ambiguous.	Medium
S03	Dead-end, high demand	253	0.02	12:00 on	Dead-end at elev 11 m, demand ~55 gpm.	Leak hidden inside a large legitimate demand.	Medium
S04	Dead-end, mid demand	131	0.02	12:00 on	Dead-end, demand ~43 gpm.	Different branch from S01; tests generalisation to another zone.	Medium
S05	Busiest 4-way hub	119	0.03	12:00 on	Degree-4, demand ~176 gpm (3rd highest).	Heavy flow through node masks the leak.	Hard
S06	Zero-demand hub	120	0.02	12:00 on	Degree-4 junction with zero demand.	Any flow loss is purely the leak; no demand noise at node.	Medium
S07	Large-pipe node	189	0.03	12:00 on	Degree-4 on the 0.76 m diameter trunk main (pipe 189).	Big pipes absorb leaks with small pressure change.	Hard
S08	Pump outlet	61	0.02	12:00 on	Node at outlet of pump 335.	Pump curve responds to the leak and distorts the pressure pattern.	Hard
S09	Near Lake pump, high elevation	10	0.01	12:00 on	Elev 44.8 m, low pressure (~16 m).	Low pressure gives low leak flow (Torricelli).	Hard
S10	Low elevation, high pressure	169	0.02	12:00 on	Elev -1.5 m, degree-4, zero demand.	High pressure drives large leak flow; strong signal.	Easy
S11	Highest elevation consumer	153	0.02	12:00 on	Elev 20.2 m, demand ~44 gpm.	Weakest pressure head among consumers.	Hard
S12	Mid-network chain node	127	0.02	12:00 on	Degree-3 inline node, elev 17.1 m.	Neighbours have similar response; Top-1 vs Top-5 gap is small.	Medium
S13	Next to Tank 1 main	40	0.02	12:00 on	Elev 40.2 m, joins 2.5 m dia. tank pipe.	Tank buffers pressure, so the signal is damped.	Hard
S14	Remote low dead-end	167	0.02	12:00 on	Elev -1.5 m, demand ~15 gpm, far end.	Far from pumps and tanks; flow signal travels a long way.	Medium

C. Same location, different leak size (node 109, highest demand, 231 gpm)

ID	Scenario	Leak node(s)	Area (m2)	Timing	Setup (how to simulate)	Why it is unique / challenge	Level
S15	Tiny seep	109	0.0005	12:00 on	Incipient leak, near sensor resolution.	Below the 0.5 m detection threshold; tests sensitivity.	Very hard
S16	Small leak	109	0.002	12:00 on	Small orifice.	Marginal detectability against demand noise.	Hard
S17	Medium leak	109	0.01	12:00 on	Typical service-line crack.	Reference size for comparing S15-S18.	Medium
S18	Pipe burst	109	0.05	12:00 on	Large orifice.	Huge pressure drop network-wide; the whole region looks affected.	Easy to detect, hard to pinpoint

D. Different timing of the leak

ID	Scenario	Leak node(s)	Area (m2)	Timing	Setup (how to simulate)	Why it is unique / challenge	Level
S19	Night-time leak	193	0.01	02:00 on	Start at 2 am, low demand, high pressure.	Minimum-night-flow condition; easiest time to see a leak.	Easy
S20	Morning-peak leak	193	0.01	08:00 on	Start during morning demand peak.	Leak coincides with a natural pressure dip.	Hard
S21	Short burst	193	0.03	14:00-15:30	Leak with end_time set, then repaired.	Event is over before the day ends; detector must catch a transient.	Medium
S22	Growing leak	193	0.002 -> 0.01 -> 0.03	06:00, 12:00, 18:00	Three stages at the same node, area grows with time.	Slowly worsening pipe crack; no sudden jump.	Hard
S23	Very late leak	193	0.02	23:00 on	Only 1-2 hourly samples after start.	Almost no post-leak data to localise from.	Hard

E. Multiple leaks at the same time

ID	Scenario	Leak node(s)	Area (m2)	Timing	Setup (how to simulate)	Why it is unique / challenge	Level
S24	Two far-apart leaks	131 and 225	0.02 each	12:00 on	Two dead-ends in different zones.	Two separate signatures; Top-K must contain both.	Medium
S25	Two adjacent leaks	171 and 173	0.02 each	12:00 on	Two ends of the 0.76 m trunk pipe 189.	Overlapping signatures can merge into one.	Hard
S26	Staggered triple leak	109, 193, 243	0.01, 0.02, 0.01	06:00, 12:00, 18:00	Three leaks start at different times.	Tests incremental detection of new leaks.	Hard
S27	Big leak masks small leak	115 and 253	0.05 and 0.003	12:00 on	One burst plus one tiny seep.	Burst dominates the score; small leak is hidden.	Very hard

F. Blocked / heavy flow (the water-flow problems in the project description)

ID	Scenario	Leak node(s)	Area (m2)	Timing	Setup (how to simulate)	Why it is unique / challenge	Level
S28	Heavy flow: hydrant flush	None (decoy at 199)	-	10:00-12:00	Add +1500 gpm demand at 199 (fire / flushing).	Large legitimate flow causes big pressure drop; must NOT be called a leak.	Medium

ID	Scenario	Leak node(s)	Area (m2)	Timing	Setup (how to simulate)	Why it is unique / challenge	Level
S29	Leak + flush decoy	Leak 121; decoy 189	0.02	12:00 on; flush 10:00-14:00	Real leak at 121 while 189 is flushed.	System must pick 121, not 189.	Very hard
S30	Blocked pipe (closed valve)	None (blockage pipe 189)	-	10:00 on	Close pipe 189 (171-173) with a control.	Downstream pressure falls like a leak; flow is rerouted. Not a leak.	Hard
S31	Leak + blocked pipe	Leak 143; block pipe 321	0.02	12:00 on; block 08:00 on	Pipe 321 (163-265) closed, leak at 143.	Blockage changes flow paths, so baseline no longer matches.	Very hard
S32	Partial blockage	None (pipe 330)	-	08:00 on	Throttle pipe 330 (60-601) with a large minor-loss factor.	Gradual flow restriction and pressure loss; no mass loss.	Hard
S33	Pump shutdown	None	-	10:00-14:00	Pump 335 set to closed.	Network-wide pressure drop; no node has mass imbalance.	Medium
S34	Leak + pump shutdown	Leak 123	0.02	12:00 on; pump off 10:00-14:00	Leak while pump 335 is off.	Pressure falls everywhere; only mass imbalance points to the leak.	Very hard

G. Operating-condition and data-quality variations

ID	Scenario	Leak node(s)	Area (m2)	Timing	Setup (how to simulate)	Why it is unique / challenge	Level
S35	Low tank levels	151	0.02	12:00 on	Start Tanks 1-3 at 20% level.	Different baseline hydraulics; baseline must be re-simulated.	Hard
S36	Heat-wave demand +30%	151	0.02	12:00 on	Global demand multiplier 1.3.	Higher flows and lower pressures everywhere.	Hard
S37	Winter demand -30%	151	0.02	12:00 on	Global demand multiplier 0.7.	Higher pressures; checks the same detector works.	Medium
S38	Noisy sensors	15	0.05	12:00 on	Gaussian noise: 0.3 m on pressure, 2% on flow.	Realistic measurement error on the S01 case.	Hard
S39	Sparse sensors	131	0.02	12:00 on	Only 20 of 92 junctions monitored.	Real utilities do not sensor every node.	Very hard
S40	Sensor dropout	253	0.02	12:00 on	Leak-adjacent sensors report a stuck value.	Faulty sensor close to the leak.	Very hard

Reference: Net3 nodes used

Node type	Net3 junctions	Why it matters
Dead-end (degree 1)	15, 35, 131, 166, 167, 203, 219, 225, 231, 243, 253	Leak affects one branch only
Hub (degree 4)	111, 115, 119, 120, 121, 169, 189, 193, 255	Leak signature spreads to four neighbours
Highest demand	109 (231 gpm), 101, 119, 151, 111, 105, 103, 199	Leak is masked by consumption
High elevation / low pressure	10, 40, 20, 50, 153, 127	Weak leak flow
Low elevation / high pressure	167, 169, 143, 171, 173, 121	Strong leak flow
Pumps	Pump 10 (Lake to node 10), pump 335 (node 60 to 61)	Pump trips and pump-outlet leaks
Trunk pipes (0.76 m)	189 (171-173), 321 (163-265), 329 (61-123), 330 (60-601), 333 (601-61)	Blockage and large-pipe cases

Note: the current detector uses a fixed 0.5 m pressure-drop threshold and compares to a clean baseline. Scenarios S15, S16, S35-S37 and S39-S40 will probably expose its limits; that is intended, and is useful material for the Semester 2 roadmap.

Implementation Roadmap

Total estimate: about 8 working days. Phases are in order; each ends with a check you can run before moving on.

Phase 1: Scenario definition (about 1 day)

• Create `simulation/scenarios.py` holding all 41 scenarios as data. Each entry has: id, name, list of leaks (node, area, start, end), list of events, global demand multiplier, noise setting, sensor mask and ground truth.
• An event is one of: hydrant, close_pipe, throttle_pipe, pump_off, tank_level.
• Keep ground truth in the same file, for example truth = ['109'], or [] for no-leak cases (S00, S28, S30, S32, S33).
• Check: the file imports and a test confirms every node and pipe ID exists in Net3.inp.

Phase 2: Extend the simulator (about 2 days)

• Add a single `run(scenario)` method to `simulation/simulator.py`. It builds a fresh network, applies everything, and runs WNTRSimulator.
• Single, multiple and staggered leaks (S01-S27): `node.add_leak(wn, area, start_time, end_time)`, once per leak.
• Growing leak (S22): three leaks at the same node with increasing areas and later start times.
• Hydrant flush (S28, S29): add a temporary extra demand at the node for the time window.
• Blocked pipe (S30, S31): close the pipe with a time control. Partial blockage (S32): raise the pipe minor loss.
• Pump trip (S33, S34): time control that closes pump 335.
• Low tanks (S35): set tank init_level. Demand +/-30% (S36, S37): scale every junction demand multiplier.
• Baselines: S35-S37 and the pump or valve events change normal operation, so each needs a matching baseline run. Decide per scenario which baseline the detector compares against.
• Convergence risk: run S18, S29 and S34 first. Large leaks combined with disturbances may fail to converge in WNTR; if so, reduce the leak area or event size and note it.
• Check: S01 and S02 reproduce the existing CSVs.

Phase 3: Dataset generation (about 1 day)

• Rewrite `generate_dataset.py` to loop over scenarios.py. Per scenario write `<id>_pressures.csv`, `<id>_flows.csv` and `<id>_meta.json` (ground truth and parameters) to data/raw_simulations/.
• Apply sensor noise (S38), sparse sensors (S39) and sensor dropout (S40) after simulation as post-processing with a fixed random seed. They need no extra simulation.
• Add a command-line option such as `--only S29` to regenerate one scenario.
• Check: 41 scenario folders or file sets exist and the meta files load.

Phase 4: Connect the backend and dashboard (about 1 day)

• Change `backend/routers/analysis_api.py` to discover scenarios from the meta files instead of a hardcoded list, and to load the matching baseline per scenario.
• Add a scenario-list endpoint returning id, name, category, level and ground truth.

• Update the dashboard scenario dropdown to group scenarios by category (A-G).
• Check: /api/analyze/<id> works for every scenario.

Phase 5: Evaluation (about 1 day)

• Extend `ml_pipeline/evaluate.py` to run all scenarios and report per scenario and per category: Top-1, Top-5, and for multi-leak cases the fraction of true leaks found.
• For no-leak scenarios report the false-alarm rate (alarm raised when none should be).
• Save the results as a CSV and a one-page summary table.
• Check: results table has 41 rows.

Phase 6: Fix what fails (about 2 days)

• Expected weak spots: tiny leaks (S15, S16) under the fixed 0.5 m threshold; decoys and blockages (S28-S34) because the detector only looks for pressure drops and flow divergence; multiple leaks (S25-S27) because it returns one ranked list; sensor problems (S38-S40).
• Possible fixes: per-node thresholds from normal variation, a blockage check (flow change with no mass loss), iterative leak removal for multiple leaks, and training a classifier on the scenario data.
• Re-run the evaluation after each change and keep the before and after table; this is the Semester 2 material.

Order of work if time is short

• Do first: S00-S14, S19-S21 (only need add_leak). Then S15-S18, S24-S27. Then S28-S34 (events). Last: S35-S40 (operating conditions and data quality).